A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.1

(2304102L)

Submitted by,

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1.1. Practice Lab Assignments

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum *p* is calculated using the formula:
p = m v

where:

p is the mass of the object (in kilograms).

m is the velocity of the object (in meters per second).

Input Format:

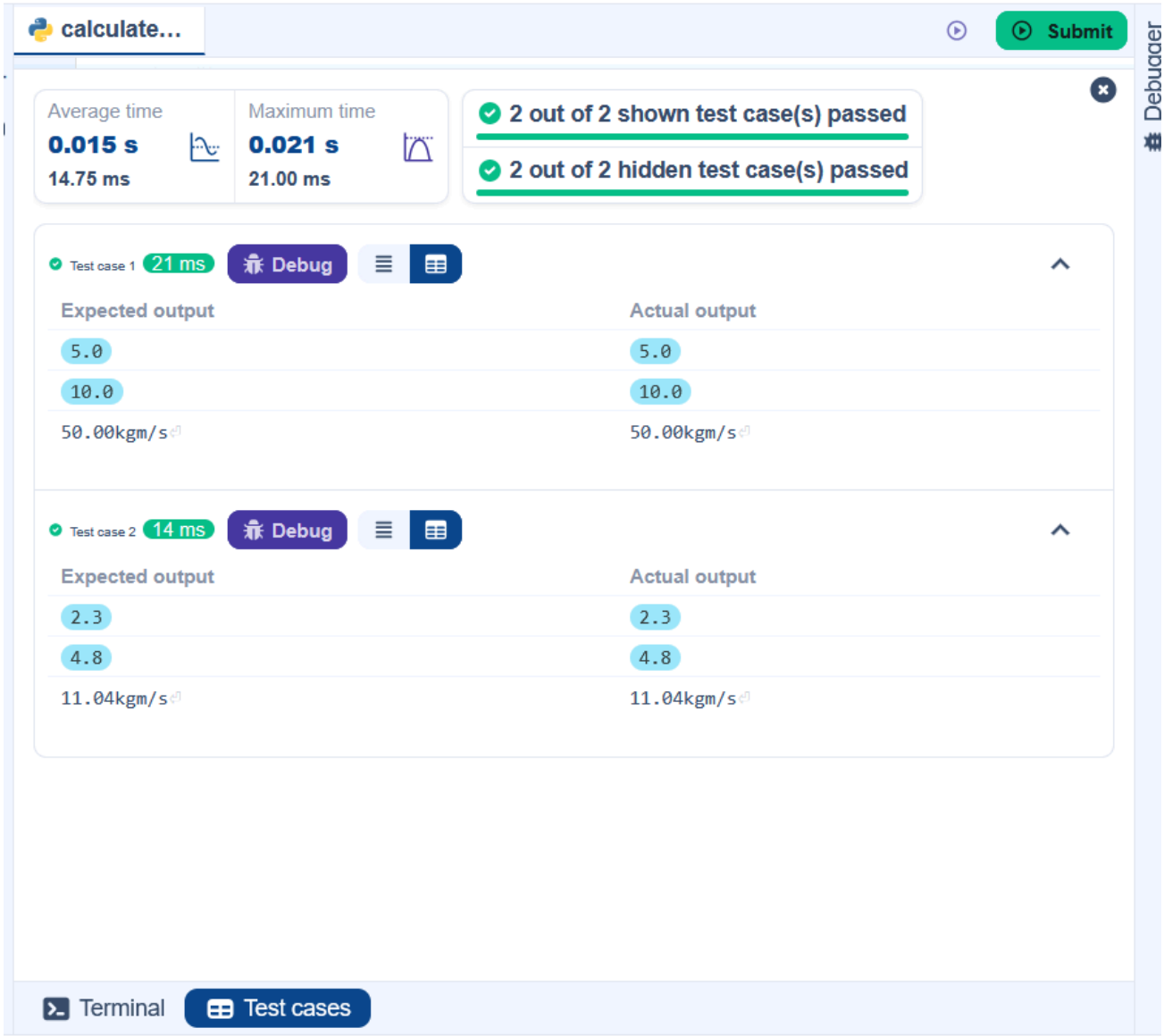
A single floating-point number representing the mass of the object in kilograms.

A single floating-point number representing the velocity of the object in meters per second.

Output Format:

The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

```
mass=float(input())
velocity=float(input())
momentum=mass*velocity
print(f"{momentum:.2f}kgm/s")
```



1.1.2 Conditional Calculation based on the Number of Digits

Write a Python program that accepts an integer as input. Depending on the number of digits in *n*.

Constraints:

$1 \leq n \leq 999$

Input Format: The input consists of a single integer *n*.

Output Format:

If *n* is a single-digit number, print its square.

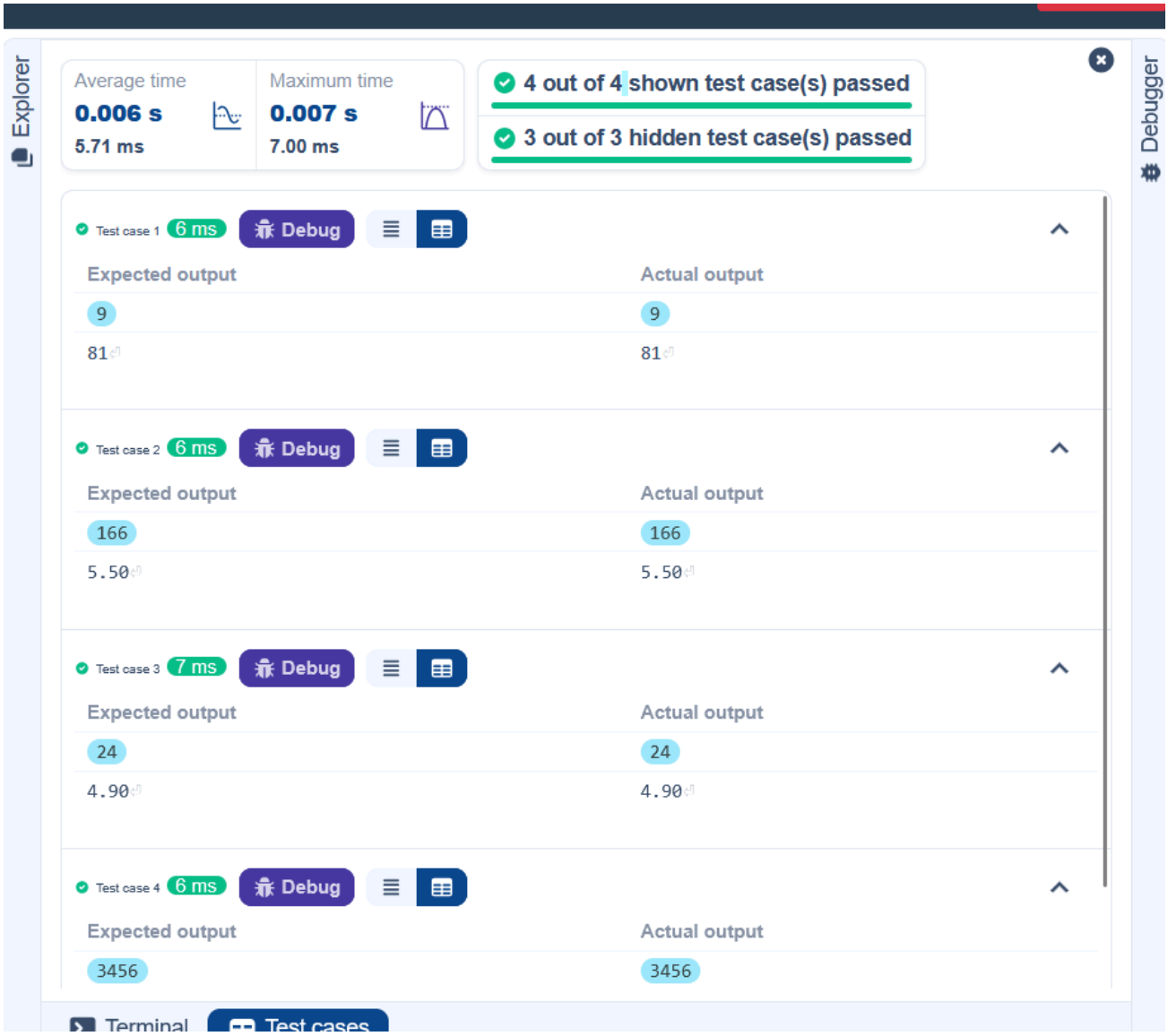
If *n* is a two-digit number, print its square root (rounded to two

decimal places).

If *n* is a three-digit number, print its cube root (rounded to two decimal places).

Else print "Invalid".

```
import math

number=int(input())

if 0<= number <=9:

        print(number**2)

elif 10<= number <=99:

        print(f"{math.sqrt(number):.2f}")

elif 100<= number <=999:

        print(f"{math.cbrt(number):.2f}")

else:

        print("Invalid")
```



1.1.3. Days Between Two Dates

Write a Python program to calculate the number of days between two dates.

Input Format:

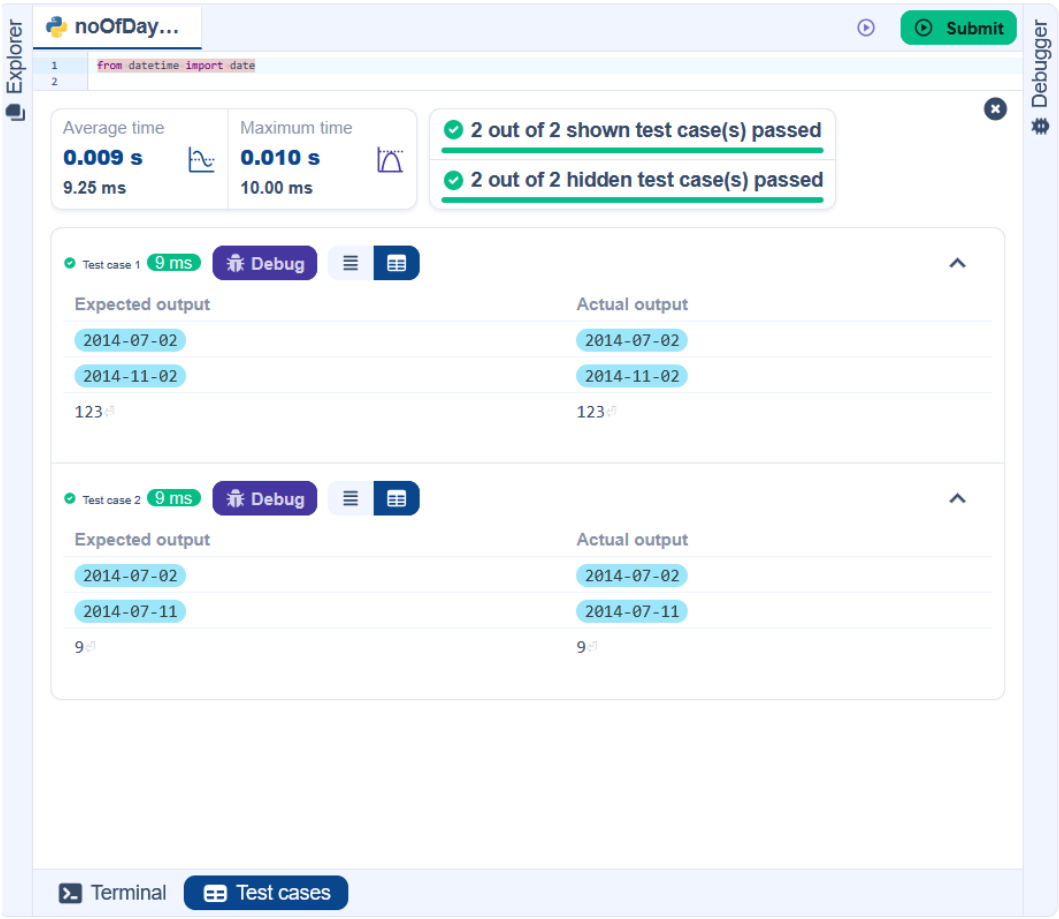
The first line contains the first date in the format *YYYY–MM–DD*.

The second line contains the second date in the format *YYYY–MM–DD*.

Output Format:

An integer representing the number of days between the given dates.

```
from datetime import datetime

date1=input().strip()
        date2=input().strip()

d1=datetime.strptime(date1, "%Y-%m-%d")
        d2=datetime.strptime(date2, "%Y-%m-%d")

difference=d2-d1
print(difference.days)
```



1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

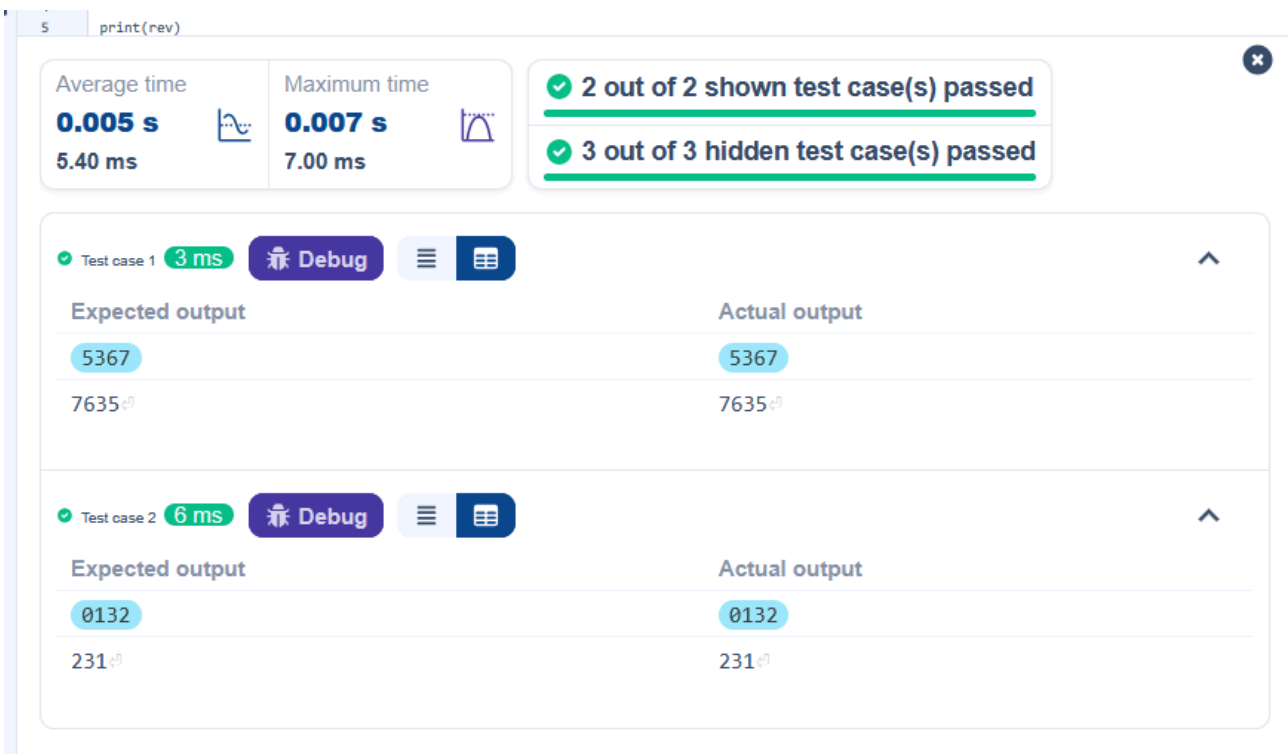
Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

```
n = int(input())

reversed_num=int(str(num)[::-1])

print(reversed_num)
```



1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number.
  <number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

```
num = int(input())

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Explorer

multiplica...

Submit

Debugger

```python
n=int(input())
for i in range(1,11):
    print(f"{n} x {i} = {n * i}")
```

Average time	Maximum time
0.004 s 4.25 ms	0.006 s 6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40
8 x 6 = 48	8 x 6 = 48
8 x 7 = 56	8 x 7 = 56
8 x 8 = 64	8 x 8 = 64
8 x 9 = 72	8 x 9 = 72
8 x 10 = 80	8 x 10 = 80

Test case 2 4 ms Debug

Expected output	Actual output
5	5
5 x 1 = 5	5 x 1 = 5
5 x 2 = 10	5 x 2 = 10
5 x 3 = 15	5 x 3 = 15
5 x 4 = 20	5 x 4 = 20
5 x 5 = 25	5 x 5 = 25
5 x 6 = 30	5 x 6 = 30
5 x 7 = 35	5 x 7 = 35
5 x 8 = 40	5 x 8 = 40
5 x 9 = 45	5 x 9 = 45
5 x 10 = 50	5 x 10 = 50

Terminal Test cases

‹ Prev Reset Submit Next ›

1.2 Lab Assignments

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer *n*, the number of courses.
- The second input will be *n* integers representing the marks of the student in each of the *n* courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

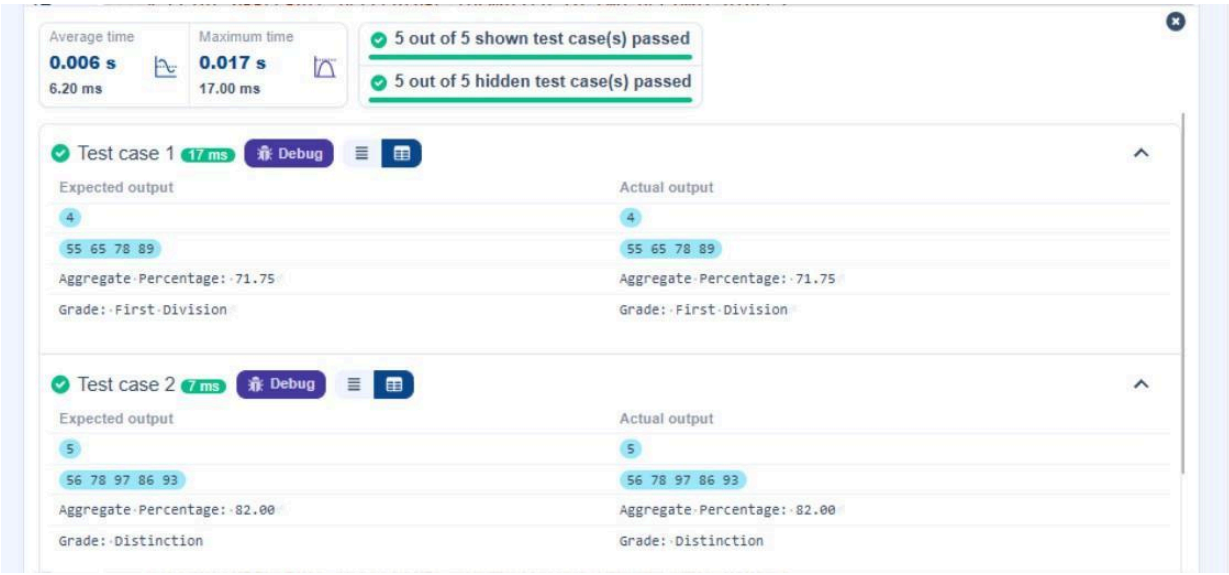
- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

```
n = int(input())
marks= list(map(int,input().split()))
failed= False
for m in marks:

    if m<40:
        failed=True
        break

if failed:
        print("Fail")
else:
        total= sum(marks)
        agg=total/n
        print("Aggregate Percentage: "f"{agg:.2f}")

        if agg > 75:
            print("Grade: Distinction")
        elif agg >=60:
            print("Grade: First Division")
        elif agg>=50:
            print("Grade: Second Division")
        elif agg >=40:
```

print("Grade: Third Division")



1.2.2. Fibonacci Series using Recursive Function

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

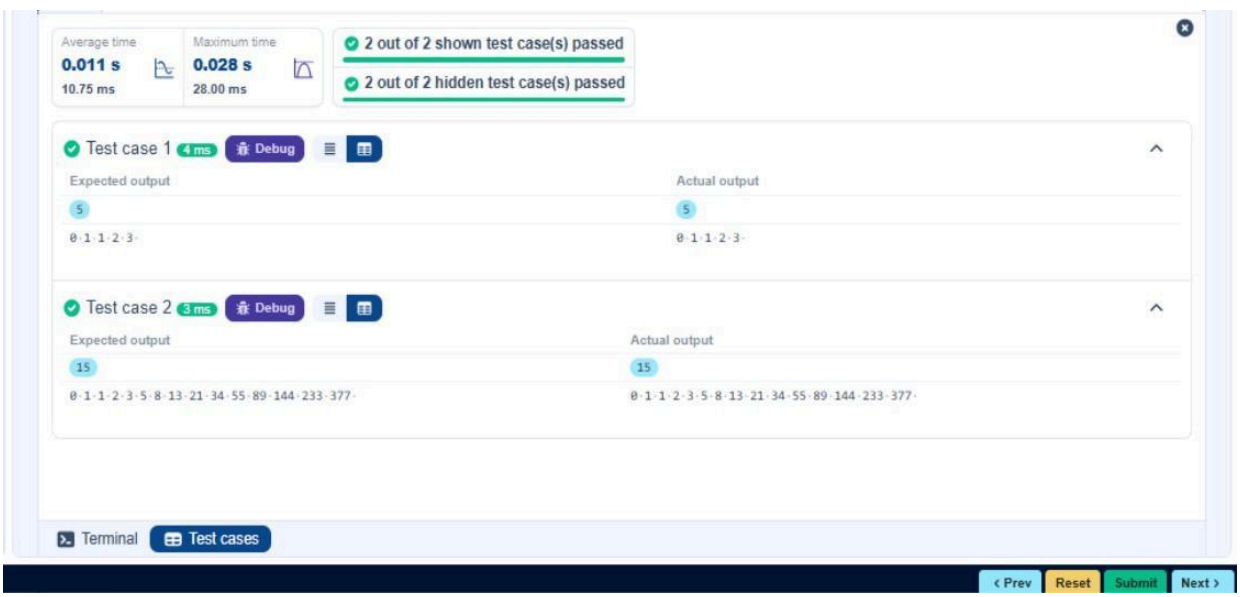
A single integer n representing the number of terms of generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

```
def fibonacci(n):
        if n==1:
                return 0
        elif n==2:
                return 1
        else:
            return fibonacci(n-1) + fibonacci(n-2)



n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```

```
def fibonacci(n):
        if n==1:
                return
        0 elif n==2:
                return 1
        else:
                return fibonacci(n-1) + fibonacci(n- 2)
n = int(input())
for i in range(1, n + 1):
        print(fibonacci(i), end=" ")
```



1.2.3. Pattern-1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Test case 1:

4

*·⏎

··⏎

··*·⏎

··*·*·⏎

```
n = int(input())
for i in range(1, n + 1):
    print('* ' * i)
```

Pattern-2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Test case 1:

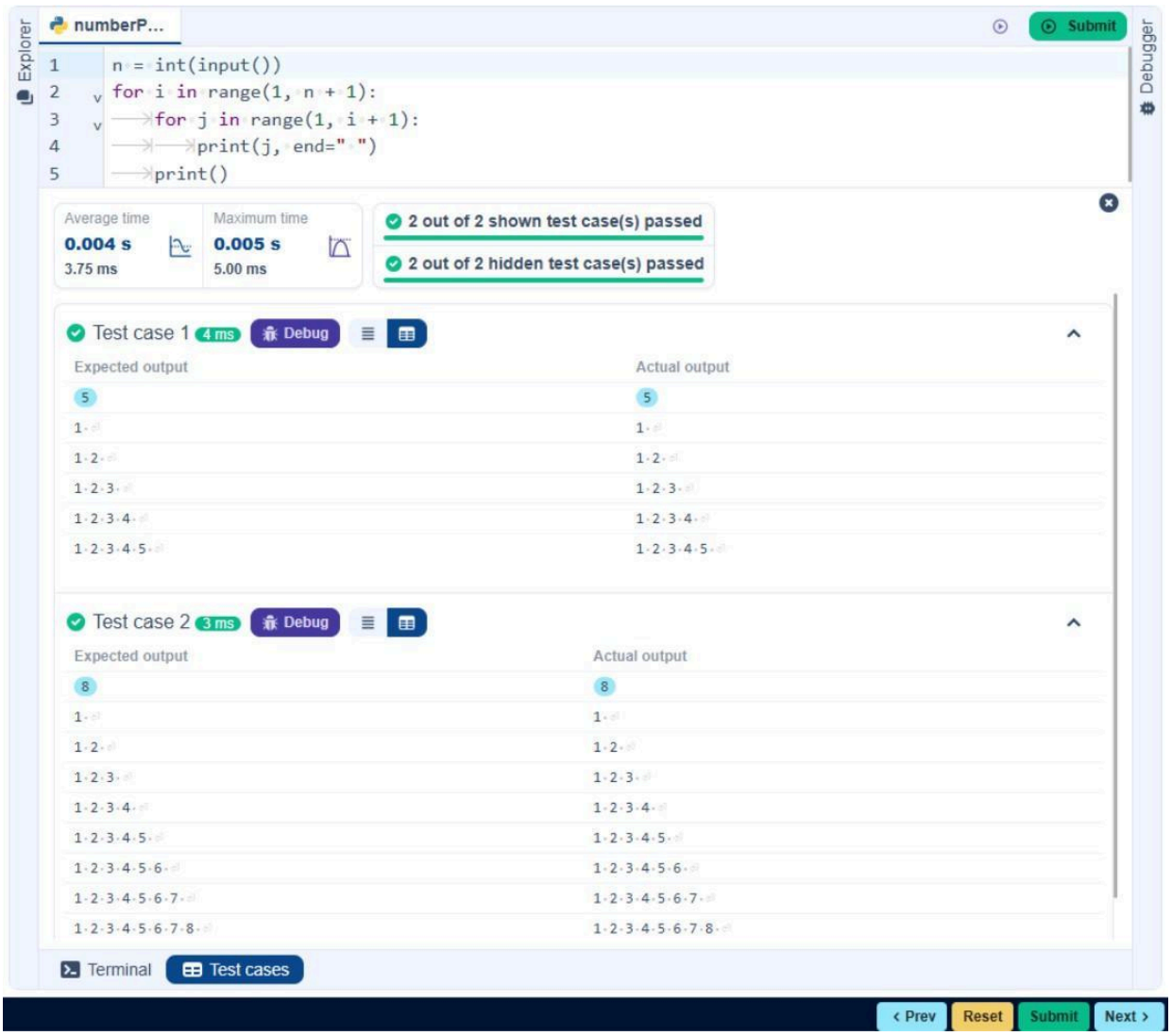
5

1·⏎

1·2·⏎

1·2·3·⏎

1·2·3·4·⏎

1·2·3·4·5·⏎

```
n = int(input())
for i in range(1, 1+n):
    for j in range(1, 1+i):
        print(j, end=" ")
    print("")
```